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History of Math

Spectrum of linear operators: A Grand Extension

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摘要

The development of spectral theory can be regarded as one of the most fascinating stories throughout the whole timeline of modern mathematics. Beginning with vibrating strings in the eighteenth century, mathematicians developed the tools of abstract operator algebras and non-commutative geometry of the twentieth and twenty-first centuries. The concept of *spectrum* then became a powerful way to help researchers understand mathematical and physical phenomena.

This paper traces the historical development of spectral theory, from the work of Sturm and Liouville in the 1830s, through the integral-equation theory of Fredholm and Hilbert, the geometric reformulation of Erhard Schmidt, the operator-theoretic axiomatisation of John von Neumann and Marshall Stone, up to the application of spectral theory in modern functional calculus, operator algebras, and applications in partial differential equations, harmonic analysis, probability, and numerical mathematics.

In each section, mathematicians who have made essential contributions shall be introduced in detail, from their life experiences to their most important papers. In this way, this paper is expected to draw a clear image as to how a intuitive physical problem extended to be one of the deepest theories of mathematics, which is what I call the great extension.

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0 Introduction

0.1 The first extension: From equation-solving to geometry

For more than a century before the modern theory of spectra of linear operators became an independent subject, the eigenvalue problem of differential and integral equations had been studied by multiple mathematicians, due to their importance in physics and engineering.

At first, solutions came differently for different types of differential equations: separation of variables for the wave equation, the method of successive approximations for Volterra equations and many more. There was no method able to solve a whole category of equations uniformly, but the need was urge.

David Hilbert made the life-saving discovery for this subject in his series of six memoirs published between 1904 and 1910. He suggested that operators could be defined and hence treated geometrically, as vectors in an infinite-dimensional Euclidean space, hence introducing what we now call **Hilbert space**. In this way, he transformed the eigenvalue problem of an integral operator as a question about the geometry of a particular sequence of vectors and projections.

His student Erhard Schmidt did more significant work in this direction in 1907, introducing the inner product, the Cauchy inequality, the Gram-Schmidt ONB construction, and the language of orthogonal projections, making the theory of abstract operators closer to what we now know as modern geometry.

0.2 The second extension: Here comes quantum

The second great extension came again from physics, which actually gave birth to this subject at the very start. Between 1925 and 1927, Heisenberg, Born, and Jordan developed matrix-mechanics, and Schrodinger in contrast built the theory of wave-mechanics around his famous Schrodinger's equation. The two ideas were eventually unified by Dirac, who raised the question of what the formalism really was. In other words, were these two theories the same or not?

The answer was given in 1929–1932 by John von Neumann. it turns out that both of these two formalisms were instances of a single mathematical structure, a self-adjoint (in general unbounded) operator on a Hilbert space, whose spectral decomposition encoded the possible outcomes of measurements and their probabilities. Therefore, **Von Neumann's spectral theorem** for unbounded self-adjoint operators, and **Marshall Stone's parallel**

theorem on one-parameter unitary groups, became the mathematical foundation of quantum mechanics.

0.3 The third extension: Algebraic structures

The story did not end there. In the 1930s and 1940s, the focus shifted from individual operators to algebras of operators. Murray and von Neumann’s rings of operators (now called **von Neumann algebras**) and **Gelfand and Naimark’s C*-algebras** together generalized the spectral theorem from a single operator to entire collections of operators. The **Gelfand–Naimark duality** between commutative C*-algebras and locally compact Hausdorff spaces revealed that the spectral theory of a commutative algebra of operators is the same thing as the topology of an underlying space. Moreover, non-commutative algebras, by analogy, are non-commutative spaces.

This insight has flowered, in the work of Alain Connes since the 1980s, into the substantial subject of *non-commutative geometry*, in which the spectral data of an operator on a Hilbert space play the role that points and metrics play in classical Riemannian geometry.

0.4 More extensions: To other branches and beyond

Spectral theory has had profound influence on various subjects: quantum mechanics as stated above, partial differential equations which has been deeply related since the origination, and also other branches of mathematics such as ergodic theory, probability theory, harmonic analysis, even number theory (via the Selberg trace formula), differential geometry (via the Atiyah–Singer index theorem), and data analysis (via principal component analysis, kernel methods, and spectral clustering).

To wrap it up, spectral theory is now widely acknowledged as one of the most powerful and essential tools in understanding the simple multiplicative actions hidden behind apparently complicated linear transformations.

0.5 What this paper is doing

This paper aims to trace the thread of the development of spectral theory, along with introducing the lives and works of major mathematicians involved: Sturm, Liouville, Fredholm, Hilbert, Schmidt, von Neumann, Stone, Gelfand, Naimark, Murray, Connes. Contents shall be organized mainly chronologically.

1 Where Did Eigenvalues Originate From?

1.1 The Vibrating String and the Birth of the Eigenvalue Problem

The first eigenvalue problem in the history of mathematics arose, somewhat indirectly, in the 18th-century debate about the vibrating string.

Imagine a string of total length L and constant linear density ρ , tied at both ends and stretched with tension T . For a small segment of the string with length Δx , the net vertical component of the tension at the two endpoints is approximately

$$T(\sin \theta|_{x+\Delta x} - \sin \theta|_x) \approx T \Delta x u_{xx}.$$

By Newton's second law applied to this segment of mass $\rho \Delta x$,

$$\rho \Delta x u_{tt} = T \Delta x u_{xx},$$

$$u_{tt} = c^2 u_{xx}, \quad c^2 = T/\rho.$$

The solution of this equation is given by the well-known method named separation of variables, dating back to Daniel Bernoulli's 1753 paper and earlier.

Assume that $u(x, t) = X(x)T(t)$, then we have

$$\frac{T''(t)}{T(t)} = c^2 \frac{X''(x)}{X(x)} = -\lambda c^2,$$

Since the left depends only on t and the right only on x , we get two ordinary differential equations

$$T''(t) + \lambda c^2 T(t) = 0, \quad X''(x) + \lambda X(x) = 0,$$

together with boundary conditions $X(0) = X(L) = 0$ coming from the fixed endpoints.

The crucial point is that the second equation does not admit a non-trivial solution for arbitrary λ , which is quite obvious. Actually, only the discrete values below permit non-trivial solutions correspondingly:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \dots$$

$$(X_n(x) = \sin(n\pi x/L))$$

These special numbers λ_n are called **eigenvalues**, and the corresponding functions X_n are the **eigenfunctions**, together an **eigenpair**. The general solution to the wave equation is then a superposition of these functions, taking form of

$$u(x, t) = \sum_{n=1}^{\infty} (a_n \cos(c\sqrt{\lambda_n} t) + b_n \sin(c\sqrt{\lambda_n} t)) \sin\left(\frac{n\pi x}{L}\right).$$

Physically, each term represents a normal mode of vibration of the string, with the frequencies $c\sqrt{\lambda_n}/2\pi$ forming the characteristic harmonic series that acoustic engineers are rather familiar with.

Now it might seem strange though, to refer to functions like $X_n(x)$ as vectors. But indeed, under geometric reinterpretation of functions as points in an infinite-dimensional vector space, this would seem natural.

For example, in the space of square-integrable (i.e. L^2) functions is an infinite-dimensional analogue of Euclidean space, it becomes natural to speak of the X_n as eigenvectors of the differential operator $-d^2/dx^2$, with inner product given by

$$\langle f, g \rangle = \int_0^L f(x) \overline{g(x)} dx$$

Moreover, the orthogonality relations

$$\int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2} \delta_{mn}$$

defines perpendicularity in this infinite-dimensional vector space, and hence the expansion of an arbitrary square-integrable function in a Fourier sine series becomes a particular case of the **resolution of the identity** in Hilbert spaces.

But of course, none of the mathematicians back then in the 18th and early 19th centuries was aware of or took use of these geometric wordings in their research upon differential equations. The first systematic and successful theoretical investigation of equations of the form $X''(x) + \lambda X(x) = 0$ and their generalizations took place around 1836–1837, carried out by the French mathematicians **Charles-François Sturm** and **Joseph Liouville**.

1.2 Sturm–Liouville Theory

The **Sturm–Liouville theory** addresses the specific series of eigenvalue problems below:

$$-\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y(x) = \lambda w(x) y(x), \quad x \in [a, b],$$

together with linear homogeneous boundary conditions at a and b , where $p, w > 0$ on $[a, b]$ and q is real-valued and continuous.

The differential operator

$$Ly = -\frac{1}{w(x)} \left(\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] - q(x)y \right)$$

is obviously formally **self-adjoint** with respect to the inner product

$$\langle f, g \rangle_w = \int_a^b f(x) \overline{g(x)} w(x) dx,$$

The self-adjointness is a crucial property ensuring the following statements:

1. The eigenvalues are all real and they form a discrete sequence $\lambda_1 < \lambda_2 < \dots$ tending to positive infinity.
2. The eigenfunctions corresponding to distinct eigenvalues are orthogonal with respect to the inner product $\langle \cdot, \cdot \rangle_w$, and that any reasonable function on $[a, b]$ can be expanded in a (generalised Fourier) series of these eigenfunctions.

1.2.1 Relevant Publications

Sturm’s earliest investigations were communicated to the Paris Academy of Sciences in September 1833 and published in summary form in Férussac’s *Bulletin des Sciences Mathématiques* the same year. The full mathematical treatment appeared in a long memoir, *Mémoire sur les équations différentielles linéaires du second ordre* (1836) and in *Mémoire sur une classe d’équations à différences partielles* (1836), both in the first volume of the *Journal de Mathématiques Pures et Appliquées*, the journal that Liouville had founded that very year.

Liouville published his own contributions in the same volume and the next, and a joint paper in 1837 cemented what is now called Sturm–Liouville theory.

1.2.2 Theory

From a modern perspective, the achievements can be summarised as follows. Sturm and Liouville proved the following theorems and results:

1. **Oscillation theorem:** the n -th eigenfunction has exactly $n - 1$ zeros in the open interval (a, b) , and the zeros of consecutive eigenfunctions interlace.
2. **Comparison theorem:** between two zeros of a non-trivial solution of one equation, every solution of a larger equation has at least one zero.
3. **The existence of an infinite sequence of eigenvalues tending to $+\infty$.**
4. **The convergence of the eigenfunction expansion of a sufficiently smooth function.**

On the other hand, in modern textbook terms, the core theorem of regular **Sturm–Liouville theory** may be stated as:

Theorem 1.1 (Regular Sturm–Liouville theorem). *For $p, w \in C^1([a, b])$ with $p, w > 0$, and $q \in C([a, b])$. Consider the eigenvalue problem*

$$-(py')' + qy = \lambda wy, \quad x \in [a, b],$$

with separated boundary conditions $\alpha_1 y(a) + \alpha_2 y'(a) = 0$, $\beta_1 y(b) + \beta_2 y'(b) = 0$, where $(\alpha_1, \alpha_2) \neq (0, 0)$, $(\beta_1, \beta_2) \neq (0, 0)$. Then:

1. *There is a sequence of real eigenvalues $\lambda_1 < \lambda_2 < \dots$ with $\lambda_n \rightarrow \infty$, each of multiplicity one.*
2. *The corresponding eigenfunctions $\{y_n\}$ form an orthogonal basis of the Hilbert space $L^2([a, b], w dx)$.*
3. *For every $f \in L^2([a, b], w dx)$,*

$$f(x) = \sum_{n=1}^{\infty} c_n y_n(x), \quad c_n = \frac{\langle f, y_n \rangle_w}{\langle y_n, y_n \rangle_w},$$

with summing convergence in L^2 .

This theorem may be regarded as the prototype of all subsequent spectral theorems. The essential features are quite clear:

1. **Self-adjointness**, which guarantees the reality of the spectrum and the orthogonality of eigenfunctions.
2. **Compactness or regularity**, which guarantees the discreteness of the spectrum.
3. **Completeness theorem**, which permits arbitrary functions to be reconstructed from the eigenfunctions.

1.2.3 Details and proof

Let's start with the first result mentioned above, one that places Sturm's work decidedly outside the framework of nineteenth-century symbolic computation, the so-called **oscillation theorem**. The theorem states that the n -th eigenfunction y_n of a regular Sturm–Liouville problem has exactly $n - 1$ zeros in the open interval (a, b) .

Sturm proved this using the second result **Sturm's comparison theorem**: if y_1 is a solution of $-(p_1 y')' + q_1 y = 0$ and y_2 is a solution of $-(p_2 y')' + q_2 y = 0$ on the same interval, with $p_1 \geq p_2 > 0$ and $q_1 \leq q_2$, then between any two consecutive zeros of y_2 there is at least one zero of y_1 .

The proof, in Sturm's hands, is a direct application of the **Wronskian determinant**: By differentiating $p_1 y_1 y_2' - p_2 y_1' y_2$ and using the two equations to eliminate second derivatives, one finds that this expression is monotone, so that it cannot change sign without forcing one of y_1, y_2 to vanish.

The argument is striking for its qualitative, geometric character at a time when most analytic results were proved by formal computation. In fact, Sturm's theorem on the zeros of linear combinations of eigenfunctions—that any non-trivial linear combination of the first n eigenfunctions has at most $n - 1$ zeros in (a, b) —is a precursor of **Courant's nodal-domain theorem** in higher dimensions. Actually, Sturm even hinted in unpublished notes at a higher-dimensional generalized result that he could not completely establish.

In addition, Liouville complemented Sturm's work by addressing the question of **eigenfunction expansion**: given a sufficiently smooth function f on $[a, b]$, can one expand f as a series $\sum c_n y_n$ of eigenfunctions, and in what sense does the series converge?

Liouville's argument, in his 1836 papers, used a clever transformation to reduce the general second-order equation to a Schrödinger-type form, after which he applied delicate convergence estimates. The convergence statements that Liouville obtained were pointwise, on functions of bounded variation, while the convergence theorems for Fourier series were proved a few decades later by Dirichlet and Jordan. It is surprising that Sturm and Liouville obtained these results in the 1830s, half a century before the advent of measure theory and the Lebesgue integral.

1.3 Lives of Charles Sturm and Joseph Liouville

Jacques Charles François Sturm was born on 29 September 1803 in Geneva to a Lutheran family of moderate means. His father, also Charles, was a teacher of arithmetic, and the family circumstances were difficult since he passed away when young Charles was only sixteen, forcing the boy to support his mother and siblings by tutoring.

Sturm entered the Geneva Academy in 1821, where his talents were noticed by Simon Lhuilier, who taught him geometry, and by Jean-Daniel Colladon, with whom Sturm collaborated on experimental research in acoustics, including a celebrated experiment which was the first reasonably accurate measurement of the speed of sound in water, made on the cold waters of Lake Geneva.

In 1824, Sturm moved to Paris as private tutor to the children of Madame de Staël, where he came into contact with the leading mathematicians of the day: Laplace, Poisson, Fourier, Ampère, Cauchy, and Arago, who introduced him to French scientific culture. He acquired French citizenship in 1833, permitting his subsequent academic appointments. He worked for some time as Arago's secretary at the *Annales de Chimie et de Physique*.

Sturm was elected to the Académie des Sciences in 1836, succeeding Ampère's mechanics chair. He became professor at the École Polytechnique in 1840 succeeding Poisson's analysis chair, and at the Sorbonne in 1857 succeeding Cauchy's mechanics chair.

Sturm's mathematical reputation was established by two pieces of work in the early 1830s. The first was his celebrated theorem on the number of real zeros of a polynomial in a given interval, presented to the Academy in 1829 and published in 1835; this earned him the Grand Prix in Mathematical Sciences in 1834, which he shared with Liouville. The second was his work on second-order linear ordinary differential equations, the first installments of which were submitted in September 1833 and which would form the basis of the theory now bearing his name.

Charles Sturm died in Paris on 15 December 1855, of severe illness exacerbated by overwork, which is quite an alert to the work-addicts nowadays. Just joking.

Joseph Liouville was born on 24 March 1809 in Saint-Omer in northern France. He was the son of a captain in the Napoleonic army.

Liouville entered the École Polytechnique in 1825 and graduated in 1827, after which he studied at the École des Ponts et Chaussées until 1830, when family illness obliged him to abandon the profession of engineer and turn to mathematical research. He became professor at the École Polytechnique in 1838, at the Collège de France in 1851, and at the Sorbonne.

Liouville is remembered today not only for his joint work with Sturm, actually, his mathematical interests were extraordinarily broad. He made foundational contributions to complex analysis, including the **Liouville's theorem** about bounded entire functions; he constructed the first explicit example of a transcendental number in 1844, which was the so-called **Liouville constant**; he contributed to Hamiltonian mechanics by proving the well-known **Liouville's theorem on phase volume**; he also set foot in fields from differential geometry to celestial mechanics, and to the theory of integration in finite terms, giving the **Liouville's theorem on integrals of elementary functions**.

Liouville also founded the *Journal de Mathématiques Pures et Appliquées* in 1836, which he edited for nearly forty years and which provided the principal venue for French mathematical publication throughout that period. However, another widespread story of his is about how his unfortunate first reading of Galois's manuscripts in 1843 leading to their belated publication in 1846.

Liouville was elected to the Académie des Sciences in 1839 and died in Paris on 8 September 1882.

The Sturm–Liouville theorems were proved in a remarkable burst of activity centred on the early issues of Liouville's new journal. Sturm published in volume 1 (1836) his *Mémoire sur les équations différentielles linéaires du second ordre* (pp. 106–186) and *Mémoire sur une classe d'équations à différences partielles*; Sturm and Liouville together published in volume 2 (1837) the joint paper *Extrait d'un Mémoire sur le développement des fonctions en séries dont les différents termes sont assujettis à satisfaire à une même équation différentielle linéaire, contenant un paramètre variable* (pp. 220–223), establishing the convergence of the eigenfunction expansion in a form analogous to Fourier series.

The intellectual partnership of these two mathematicians, both still in their thirties at the time, illustrates the dynamic state of French mathematics in those years. Although Sturm and Liouville published only one paper jointly, their separate papers from this period form a coherent body of work, and the close interaction between them, attested by their correspondence, can be regarded as one of the earliest examples of a research programme in modern mathematics.

1.4 Weyl's 1910 Paper: From Sturm–Liouville to Singular Problems

The Sturm–Liouville theory described above applies to a regular interval, which leads natural interest on the **singular case**, where the interval is unbounded or the coefficients

have singularities at the endpoints. The first systematic treatment of the singular case was given by **Hermann Weyl** in a celebrated paper, *Über gewöhnliche Differentialgleichungen mit Singularitäten und die zugehörigen Entwicklungen willkürlicher Funktionen*, published in *Mathematische Annalen* 68 (1910), pp. 220–269.

1.4.1 Details and proof

Weyl distinguished, at each singular endpoint, between the **limit-point** case, where the equation $(pu')' + qu = \lambda u$ has only one square-integrable solution near the endpoint for non-real λ , and the **limit-circle** case, where all solutions are square-integrable. To be specific, in the limit-point case, no boundary condition is needed at the singular endpoint to obtain a self-adjoint operator; in the limit-circle case, a boundary condition is needed and the choice generates a one-parameter family of self-adjoint extensions.

Weyl’s analysis brought into spectral theory, for the first time, the concepts of **continuous spectrum** and **singular continuous spectrum**, while in the regular case, in contrast, the spectrum is purely discrete. Weyl’s paper is also notable for its careful use of complex analysis to construct the resolvent. It thus bounds, on the one hand, to the explicit calculations of the older differential-equations tradition, and pointing, on the other, toward the operator-theoretic abstractions of the next generation.

1.4.2 Life of Hermann Weyl

Weyl was a remarkable figure whose career spanned an enormous range of mathematics and mathematical physics. Born in 1885 at Elmshorn, near Hamburg, he completed his doctorate at Göttingen in 1908 under Hilbert. His doctorate thesis was on the theory of singular integral equations, and his 1910 paper grew out of these investigations.

Weyl’s later work touched on practically every major topic in twentieth-century mathematics:

1. The **equidistribution theorem** of analytic number theory in 1916.
2. The **gauge-theoretic unification** of electromagnetism and gravitation in 1918.
3. **The Peter–Weyl theorem** in 1927, which is an important result in the representation theory of compact Lie groups.
4. The foundations of **relativity** in his book *Raum, Zeit, Materie* in 1918, which went through five editions.

5. The quantum mechanics of the spinning electron, where the so-called **Weyl spinors** and the **Weyl equation** appeared.
6. The philosophy of mathematics, where his book *Das Kontinuum* in 1918 gave a predicative reconstruction of analysis.

Moreover, his 1928 book *Gruppentheorie und Quantenmechanik* was, alongside von Neumann's later monograph, one of the seminal early accounts of the mathematical structure of quantum mechanics.

Weyl emigrated from Germany in 1933, joined the Institute for Advanced Study at Princeton, and remained there for the rest of his career. He died in 1955 at Zürich.

1.4.3 Summary

In summary, the theory developed by Sturm and Liouville treated only what is today called the regular case: a finite interval, with positive smooth coefficients. The natural extension to singular problems—infinite intervals, vanishing or singular coefficients—was a much more delicate matter.

Hermann Weyl, in his celebrated 1910 paper, gave the first systematic treatment of this case, introducing the *limit-point* versus *limit-circle* classification, and showed that depending on this, the spectrum could be either purely discrete or contain a continuous part. The full understanding of singular Sturm–Liouville problems would have to wait until the 1930s, with the work of E. C. Titchmarsh, M. H. Stone, K. O. Friedrichs, and others.

By his demonstration that there are differential operators with continuous spectrum, Weyl forced the mathematical community to confront the fact that the comfortable picture of an orthogonal basis of eigenfunctions, valid in the Sturm–Liouville case, breaks down in general. Some new and more flexible notion of **spectrum** was needed.

1.5 Integral Equations: Volterra, Fredholm, and the Setting for Hilbert

A second strand of development, parallel to and eventually intertwined with Sturm–Liouville theory, was the theory of **integral equations**.

1.5.1 Vito Volterra

Although integral equations had been considered sporadically since at least Abel's 1826 paper on the tautochrone problem, and although Carl Neumann had introduced what

is now called the Neumann series in his 1877 work on potential theory, it was the Italian mathematician **Vito Volterra** who first systematically studied them. In a series of papers from 1896 to 1900, Volterra considered integral equations which are now called **Volterra integral equations**:

$$\phi(x) = f(x) + \lambda \int_a^x K(x, t)\phi(t) dt,$$

He proved that for continuous kernel K , such equations always have a unique solution given by an absolutely convergent series. Although elegant, the result was relatively limited.

1.5.2 Erik Ivar Fredholm

The decisive step was taken by **Erik Ivar Fredholm** in 1900–1903. Fredholm considered integral equations of the form

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t)\phi(t) dt,$$

with fixed limits, and showed that the **theory of solvability** of such equations are similar to that of the theory of finite linear systems. This is what we now call **Fredholm alternative theorem**.

Explicitly, the eigenvalues of K are the values of $1/\lambda$ for which the homogeneous equation has a non-trivial solution, while for all other λ , the equation has a unique solution. Moreover, the inhomogeneous equation is solvable if and only if f is orthogonal to all solutions of the adjoint homogeneous equation.

The first announcement was made in a brief paper of 1900 in the *Comptes Rendus* of the Royal Swedish Academy, but the full development came in the celebrated 1903 paper *Sur une classe d'équations fonctionnelles*, published in *Acta Mathematica* 27, pp. 365–390. This paper is also regarded as the one of the most important foundations of operator theory, also paving the way for Hilbert's research in a few decades to come.

2 The Existence and Uniqueness of Eigenpairs: Fredholm, Hilbert, Schmidt

2.1 Erik Ivar Fredholm and the 1903 Paper

2.1.1 The life of Erik Ivar Fredholm

Erik Ivar Fredholm was born on 7 April 1866 in Stockholm. He was the son of a wealthy merchant who made his fortune by replacing gas lamps with electric ones.

Fredholm was educated at the Beskowska School in Stockholm, where he performed excellently. After his *baccalauréat* (French for high-school graduation) in 1885, he entered the Royal Technological Institute in Stockholm, and later transferred to the University of Uppsala.

Fredholm took his master's degree in 1888 and his doctorate in 1898. His doctorate thesis was directed by Mittag-Leffler, and focused on partial differential equations associated with the deformation of an anisotropic elastic medium.

In 1898, Fredholm spent some time in Paris in 1899, where he met mathematicians including Picard, Hadamard, and Poincaré. His contacts with these researchers would later prove important for the dissemination of his work. From 1898, Fredholm worked as a lecturer in mathematical physics at Stockholm University, and from 1906 he was full professor of rational mechanics and mathematical physics there.

Fredholm had many interests outside mathematics. As a starter, he was a skilled craftsman who built his own violins from half a coconut shell. He also was able to construct mechanical computers. In addition, he served as an actuary at the Skandia Insurance Company from 1904 to 1907, applying his mathematical skills to insurance problems.

At the time of his death in 1927, he was working on the mathematics of violin acoustics. Yet it's a pity that his unfinished manuscripts on the subject were never deciphered.

2.1.2 The 1902 paper

Fredholm's seminal 1903 paper *Sur une classe d'équations fonctionnelles* considered the integral equation

$$\phi(x) - \lambda \int_a^b K(x, t) \phi(t) dt = f(x),$$

in which K is a continuous kernel on $[a, b] \times [a, b]$ and f is continuous on $[a, b]$.

By analogy with the formula for the determinant of a finite matrix, in a quite genius

way, Fredholm introduced what is now called the **Fredholm determinant** for operators:

$$D(\lambda) = \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!} \int_a^b \cdots \int_a^b \det [K(x_i, x_j)]_{i,j=1}^n dx_1 \cdots dx_n,$$

together with a corresponding numerator (the so-called **first Fredholm minor**).

Fredholm showed that the **resolvent** of K , i.e. the kernel of the inverse operator—can be expressed as a quotient of these two entire functions of λ . The values of λ for which the homogeneous equation ($f=0$) has a non-trivial solution are precisely the zeros of $D(\lambda)$.

The completeness of this theory is striking: the entire theory of finite linear systems, including rank, kernel, image, alternative for solvability of the inhomogeneous problem, transfers to the infinite-dimensional setting of integral equations. The theorems together are now known as the **Fredholm alternative**:

Theorem 2.1 (Fredholm alternative). *Let $K \in L^2([a, b]^2)$ and consider the integral equation $(I - \lambda K)\phi = f$ on $L^2([a, b])$, with adjoint equation $(I - \bar{\lambda}K^*)\psi = 0$. Exactly one of the following holds:*

1. *For every f , the equation $(I - \lambda K)\phi = f$ has a unique solution. The homogeneous equation has only $\phi = 0$, and the same is true for the adjoint.*
2. *The homogeneous equations $(I - \lambda K)\phi = 0$ and $(I - \bar{\lambda}K^*)\psi = 0$ both have non-trivial solutions, both with the same finite dimension.*

In this case, the inhomogeneous equation is solvable if and only if f is orthogonal to every solution of the adjoint homogeneous equation.

Fredholm's paper was striking news for many. Erik Holmgren delivered lectures on Fredholm's results in Göttingen in 1901, and Hilbert immediately recognised their importance. Poincaré, in a report to the Paris Academy, used the methods in Fredholm's work in his own studies of celestial mechanics.

In 1903, Fredholm received the Wallmark Prize from the Royal Swedish Academy of Sciences, and the Poncelet Prize from the French Academy of Sciences in 1908, and also an honorary doctorate from Leipzig in 1909.

2.2 David Hilbert: Six Memoirs and the Geometric Vision

2.2.1 Life of David Hilbert

David Hilbert was already, by the time of Fredholm's paper, one of the most prominent mathematician in Germany and arguably in the world.

Hilbert was born on 23 January 1862 in Königsberg. He was the son of a county judge. He received his early education at the Friedrichskolleg and later at the University of Königsberg, taking his doctorate in 1885 under Ferdinand von Lindemann.

After habilitation in 1886, Hilbert remained at Königsberg as Privatdozent until 1895, when, on Klein's recommendation, he moved to a chair at Göttingen, where he would remain for the rest of his life.

By 1900, Hilbert has become extremely famous, both for his work on invariant theory, algebraic number theory, and the foundations of geometry, and for his celebrated lecture at the Paris International Congress where he proposed the twenty-three problems that would shape the mathematics of the twentieth century.

Hilbert and Göttingen had a strong relation in between. As acknowledged by many mathematicians, the Göttingen of 1900 to 1933 was perhaps the center of mathematical in the world. Felix Klein, who took up the chair of mathematics at Göttingen in 1886, immediately begun the work of transforming the university into a research institute of quality no one can align with. Hilbert's appointment in 1895, on Klein's recommendation, was a key step in this programme.

Actually, by the early years of the twentieth century, Göttingen had attracted not only Hilbert and Klein but also many other distinguished mathematicians such as Hermann Minkowski, Edmund Landau, Carl Runge, Constantin Carathéodory. It also taught multiple doctoral students who later made great achievements: Otto Blumenthal (1898), Felix Bernstein (1901), Hermann Weyl (1908), Erhard Schmidt (1905), Richard Courant (1910), Erich Hecke (1910), Hugo Steinhaus (1911), Wilhelm Ackermann (1925), and many others.

The combination of a great senior figure with a steady stream of brilliant students made Göttingen an extremely suitable environment for creation. Great amounts of important work on integral equations between 1904 and 1910 was done. Sadly, the Nazi purge of 1933 destroyed this environment in a matter of months: Weyl, Courant, Bernstein, Landau, Noether, and many others were forced to leave, and the dominant German mathematical centre of the previous half-century ceased to exist. Hilbert implied at a banquet by the new minister of education Mathematics in Göttingen is really not there any more.

2.2.2 Hilbert's work on integral equations

Beginning around 1902, Hilbert immersed himself in the theory of integral equations inspired by Fredholm's work. His investigations were published as a series of six memoirs

in the *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen* between 1904 and 1910, and were collected in 1912 as the book *Grundzüge einer allgemeinen Theorie der linearen Integralgleichungen* (*Foundations of a General Theory of Linear Integral Equations*).

In the first memoir, Hilbert extended Fredholm's results to the case of **symmetric kernels**, that is, kernels satisfying $K(x, t) = K(t, x)$. For such kernels Hilbert proved that the eigenvalues are all real, that the corresponding eigenfunctions can be chosen to form an orthonormal system, and that one has the expansion theorem and the bilinear formula

$$K(x, t) = \sum_n \frac{1}{\lambda_n} y_n(x) \overline{y_n(t)},$$

where $\{y_n\}$ is the orthonormal system of eigenfunctions and $\{\lambda_n\}$ the corresponding eigenvalues, and the convergence of sum holds in an appropriate sense. This already amounts to a spectral theorem in the symmetric, finite-rank or compact, case.

What is much more remarkable, however, is what Hilbert did in his fourth memoir of 1906. He showed that bilinear forms, or infinite linear systems in the form of

$$y_i = \sum_{j=1}^{\infty} a_{ij} x_j, \quad i = 1, 2, \dots,$$

where the matrix $A = (a_{ij})$ is symmetric and the vectors (x_i) lie in the space ℓ^2 , can be treated by analogous methods to that above. Thus one obtains analogous expansion theorems.

This is when the space ℓ^2 made its first explicit appearance in Hilbert's work in this paper, although a fun fact is that Hilbert himself never used the term *space*. Instead, he spoke of the totality of square-summable number sequences.

The geometric language was introduced shortly afterwards by Schmidt and by Riesz.

A novel feature of Hilbert's treatment, again from a modern perspective, was his discovery that not all bilinear forms admit a discrete spectral decomposition. Some have a *Streckenspektrum* (German for **continuous spectrum**), and Hilbert succeeded in giving a representation of arbitrary bounded symmetric bilinear forms in terms of an integral over the spectral measure. Despite flaws in the mathematical language Hilbert chose to use, this 1906 result is essentially the first **continuous spectral theorem** in mathematics.

The word **spectrum** itself, was in fact Hilbert's own coinage coming from physical phenomena. Hilbert chose the word *Spektrum* by analogy with the optical spectrum

of light. This is natural to see, since a prism resolves a beam of white light into its monochromatic components, just as the spectral decomposition of an operator resolves it into its eigenvalues or its spectral density.

This intuition actually made sense. By the late 1920s, the spectral lines of atoms would be understood through quantum mechanics, as the eigenvalues of the Hamiltonian operator of the atom.

2.3 The Geometric Vision of Erhard Schmidt

As mentioned in the previous section, Hilbert's investigations of integral equations were somehow difficult to read. His proofs proceeded by limiting arguments from the finite-dimensional case, requiring at every step delicate convergence estimates, and the role of the symmetry of the kernel was somewhat obscured by these technicalities. The decisive simplification was due to Hilbert's student Erhard Schmidt.

2.3.1 Life of Erhard Schmidt

Erhard Schmidt was born on 13 January 1876 in Dorpat, in present-day Estonia, then part of the Russian Empire. His father Alexander Schmidt was a medical biologist, and the family was Baltic German.

Schmidt's university education followed the typical German pattern of his time, studying at several different universities. He first attended his local university in Dorpat, and later went to Berlin, where he studied with Hermann Schwarz. Finally he went to Göttingen, where he became a doctoral candidate under Hilbert.

Schmidt took his doctorate in 1905 with a dissertation entitled *Entwicklung willkürlicher Funktionen nach Systemen vorgeschriebener* (*Expansion of Arbitrary Functions in Terms of Prescribed Systems*), the main results of which were published in his celebrated 1907 paper.

After habilitation at Bonn in 1906, Schmidt held positions at Zürich, Erlangen, and Breslau before being appointed to a professorship at the University of Berlin in 1917, where he succeeded H. A. Schwarz. He remained at Berlin for the rest of his career and died there on 6 December 1959.

2.3.2 The 1907 paper

Schmidt's most famous papers from this period, the two-part *Zur Theorie der linearen und nichtlinearen Integralgleichungen* (*On the Theory of Linear and Nonlinear Integral Equations*), appeared in *Mathematische Annalen* volumes 63 (1907), pp. 433–476, and 64

(1907), pp. 161–174. These papers reexplained Hilbert’s theorems on integral equations with symmetric kernel while avoiding limiting arguments from finite-dimensional approximations, by using geometric methods.

Suppose K is a continuous symmetric kernel on $[a, b]^2$ and consider the operator on the space of continuous functions on $[a, b]$:

$$T\phi(x) = \int_a^b K(x, t)\phi(t) dt$$

The operator is symmetric, in the sense that $\langle T\phi, \psi \rangle = \langle \phi, T\psi \rangle$, and bounded. To construct an eigenfunction, Schmidt observed that **if the function $\phi \mapsto \langle T\phi, \phi \rangle / \|\phi\|^2$, defined on the unit ball, attains its maximum on a function ϕ_1 , then this maximizer is an eigenfunction with eigenvalue equal to that maximum value.**

Once ϕ_1 has been found, one passes to the orthogonal complement and repeats the process, finally obtaining an orthonormal sequence of eigenfunctions $\{\phi_n\}$ with eigenvalues $\{\lambda_n\}$. The eigenvalues decrease to zero in absolute value, since K is square-integrable and hence T is compact, the eigenvalues cannot accumulate at any non-zero value. The expansion theorem then follows from the observation that the orthogonal complement of the closed span of $\{\phi_n\}$ lies in the kernel of T .

This argument makes the last piece of puzzle for the **Hilbert–Schmidt spectral theorem** for compact self-adjoint operators, which is undoubtedly the foundational result of all spectral theory.

Theorem 2.2 (Hilbert–Schmidt spectral theorem). *Let H be a separable Hilbert space and $T : H \rightarrow H$ a compact self-adjoint operator. Then there exist a (possibly finite) orthonormal sequence $\{e_n\}$ in H and a sequence of real numbers $\{\lambda_n\}$, with $\lambda_n \rightarrow 0$ if the sequence is infinite, such that*

$$Tx = \sum_n \lambda_n \langle x, e_n \rangle e_n \quad \text{for every } x \in H.$$

The non-zero λ_n are the non-zero eigenvalues of T , and e_n is a corresponding orthonormal basis of eigenvectors.

2.3.3 Details and proof

Schmidt’s proof proceeds in three steps:

1. Show that $\|T\| = \sup_{\|x\|=1} |\langle Tx, x \rangle|$.

2. Use compactness to show that this supremum is attained on some unit vector e_1 , which is an eigenvector with eigenvalue $\pm\|T\|$.
3. Pass to $e_1^\perp$, which is invariant under T , and iterate.

The argument requires nothing more than the Cauchy–Schwarz inequality, the parallelogram law, and the definition of compactness.

Explicitly, for self-adjoint T , the quantity $M = \sup_{\|x\|=1} \langle Tx, x \rangle$ and $m = \inf_{\|x\|=1} \langle Tx, x \rangle$ satisfy $\|T\| = \max(|m|, |M|)$. Assume w.l.o.g. by changing T to $-T$ if necessary, that $\|T\| = M$.

Choose a sequence of unit vectors x_n with $\langle Tx_n, x_n \rangle \rightarrow M$. By compactness, Tx_n has a convergent subsequence, so w.l.o.g. write $Tx_n \rightarrow y$. By

$$\|Tx_n - Mx_n\|^2 = \|Tx_n\|^2 - 2M\langle Tx_n, x_n \rangle + M^2$$

and

$$\|Tx_n\|^2 \leq \|T\| \langle Tx_n, Tx_n / \|Tx_n\| \rangle \leq \|T\|^2 = M^2$$

we have

$$\|Tx_n - Mx_n\|^2 \rightarrow 0$$

which means that $Mx_n \rightarrow y$ and $x_n \rightarrow y/M$.

The limit $e_1 = y/M$ is then a eigenvector with $Te_1 = Me_1$.

Restrict T to the orthogonal complement $e_1^\perp$, which is invariant by self-adjointness, and repeat the argument. Thus one obtains an orthonormal sequence $e_1, e_2, \dots$ of eigenvectors with eigenvalues $\lambda_1, \lambda_2, \dots$ of decreasing absolute value.

The fact that $\lambda_n \rightarrow 0$ can be proved by contradiction: if not, then $\{e_n\}$ is bounded but $Te_n = \lambda_n e_n$ has no convergent subsequence, contradicting the compactness of T .

Finally, on the closure of the span of $\{e_n\}$, the operator T is given by the convergent series, so on the orthogonal complement of this span, T is zero. This completes the proof.

This argument is a lot simpler in contrast to Hilbert's original treatment, which proceeded by approximating the integral operator by finite-dimensional matrices and passing to the limit. Hence unsurprisingly, Schmidt's geometric proof became an important prototype of modern approaches to spectral theory. People now focus more on the abstract structural features of the operator, such as self-adjointness and compactness.

Moreover, Schmidt's ideas also exhibits the variational characterisation of eigenvalues

that lies behind the much later *minimax principle* of Courant and Fischer:

$$\lambda_n = \min_{\substack{V \subset H \\ \dim V = n}} \max_{\substack{x \in V \\ \|x\|=1}} \langle Tx, x \rangle = \max_{\substack{V \subset H \\ \dim V = n}} \min_{\substack{x \in V \\ \|x\|=1}} \langle Tx, x \rangle$$

for a compact self-adjoint operator with positive eigenvalues arranged in decreasing order. The minimax principle, in turn, is the foundation of the modern theory of eigenvalue inequalities and of variational methods in PDE.

2.4 Schmidt's Other Contributions and the Birth of Hilbert Space

Schmidt's 1907 paper contained more than the proof of the Hilbert–Schmidt theorem. In its second part, Schmidt also developed what is now called the **singular value decomposition** of a non-self-adjoint compact operator, anticipating much later developments. He proved that for a compact operator T one can write

$$Tx = \sum_n s_n(T) \langle x, v_n \rangle u_n,$$

where $\{u_n\}$ and $\{v_n\}$ are orthonormal sequences and the $s_n(T) > 0$ are the so-called **singular values**, the eigenvalues of $(T^*T)^{1/2}$.

A year later, in his 1908 paper *Über die Auflösung linearer Gleichungen mit unendlich vielen Unbekannten* (*On the Solution of Linear Equations with Infinitely Many Unknowns*), *Rendiconti del Circolo Matematico di Palermo* 25, pp. 53–77, Schmidt defined explicitly the geometric structure of ℓ^2 . That is to say, he spoke of **length** (norm), **angle**, **orthogonality**, and **projection**, and gave what is now called the **Gram–Schmidt orthonormalization process**, although the same process had been described, less explicitly, by J. P. Gram in 1879 and even earlier by Laplace and Cauchy.

This 1908 paper is widely regarded as the birth certificate of the abstract notion of Hilbert space, although the specific term *Hilbertraum* was not used until somewhat later, and the truly axiomatic definition would come only with von Neumann in 1929–1930.

After moving to Berlin, Schmidt's interests turned to topology, where he gave a new proof of the **Jordan curve theorem**. He also worked on isoperimetric inequalities, but his work in functional analysis remained his most influential contribution.

2.5 The Riesz–Schauder Theory and the Notion of Compactness

A decade after Schmidt's papers were published, the abstract notion of **compact operator** emerged, due primarily to the Hungarian mathematician **Frigyies (Frédéric) Riesz**. In his 1916 paper *Über lineare Funktionalgleichungen*, *Acta Mathematica* 41, Riesz proved that the Fredholm alternative continues to hold for any compact linear operator on a Banach space. As a remark, the concept of Banach space was not yet abstracted at that time, but treated implicitly through the concrete spaces $C[a, b]$ and L^p .

In 1930, **Juliusz Schauder** extended these results. The combined **Riesz–Schauder** theory yields the following statement:

Theorem 2.3 (Riesz–Schauder). *Let T be a compact linear operator on a Banach space X and let $\lambda \neq 0$. Then the operator $T - \lambda I$ has finite-dimensional kernel and closed range of finite codimension equal to the dimension of the kernel of the adjoint $T^* - \bar{\lambda}I^*$. The non-zero spectrum of T consists of countably many eigenvalues of finite multiplicity, with no accumulation point other than zero.*

It can be seen easily that the diagonalization of matrices is a special case of the Hilbert-Schmidt theory, i.e. the spectral theory of compact self-adjoint operators.

3 Quantum Mechanics and the Reorientation of Spectral Theory

3.1 The Crisis of Old Quantum Theory

By the mid-1920s, the old quantum theory developed by Bohr and Sommerfeld was in crisis. Their model succeeded in explaining the hydrogen spectrum, the photoelectric effect, and certain features of atomic physics, but it had no rigorous theoretical foundation, and failed conspicuously to explain even the helium spectrum.

From 1925 to 1927, in a rather surprising burst of creation, two essentially different formalisms were proposed: **matrix mechanics**, due to Heisenberg, Born, and Jordan; and **wave mechanics**, due to Schrödinger, building on de Broglie.

Heisenberg's matrices and Schrödinger's differential operator both satisfied the canonical commutation relation $[Q, P] = i\hbar$, but their relationship was unclear, and the mathematical status of either formalism was uncertain. Schrödinger himself, in March 1926, proved that the two theories yielded equivalent predictions, but a fully rigorous demonstration of their equivalence was still missing.

Yet one thing about this new physics was clear: **physical observables, such as position, momentum, energy, and angular momentum, were to be represented by self-adjoint operators on a Hilbert space, and the values that could be observed in measurement were the spectrum of the corresponding operator.**

Several questions immediately arose.

1. What is the precise sense in which Heisenberg's infinite matrices and Schrödinger's differential operators are equivalent?
2. How is one to make sense of the spectral decomposition of an unbounded operator like the position operator Q , whose spectrum is the entire real line and which has no eigenvectors in the usual sense?
3. What is the meaning of an exponentiated Hermitian operator $\exp(i\hbar H/\hbar)$, and how is one to interpret the time-dependent Schrödinger equation rigorously?

These questions defined the agenda for a generation of mathematical physicists, among which are **John von Neumann and Marshall Stone**, working independently but exchanging ideas, between 1927 and 1932, and made one the most significant breakthroughs.

3.2 John von Neumann: Life and Mathematical Background

John von Neumann was born on 28 December 1903 in Budapest. He was the eldest of three sons of Max Neumann, a banker who was elevated to the Hungarian nobility in 1913. The family was wealthy and assimilated Jewish, and the young János received a private education from tutors before entering the Lutheran Gymnasium, one of Budapest's most prestigious schools, where his classmates included Eugene Wigner.

From the very beginning, Neumann's mathematical talent was prodigious. He could joke in Classical Greek by the age of eight, memorise pages of the telephone book, and divide eight-digit numbers in his head. By the time he entered university, he had already published two mathematical papers; the second, of 1923, gave the modern definition of an ordinal number as the set of all smaller ordinal numbers.

Von Neumann's father, fearing that mathematics would not provide a stable career, urged him to study chemistry. As a compromise, von Neumann simultaneously pursued chemical engineering at the ETH in Zürich, achieving a degree in 1925, and studied mathematics at the University of Budapest, receiving a doctorate in 1926. His doctorate thesis was on the axiomatisation of set theory under the formal direction of L. Fejér and L. Kürschák.

For the academic year 1926–1927, Neumann held a Rockefeller fellowship at Göttingen, working under Hilbert. From 1927 to 1929 he was *Privatdozent* at the University of Berlin. In Berlin he continued his work on the foundations of mathematics, on operator theory, and on the mathematical foundations of quantum mechanics.

In 1930, Neumann received an invitation to lecture at Princeton University, in 1931 he became a professor there, and in 1933 he was appointed one of the original six professors at the newly founded and now distinguished Institute for Advanced Study, alongside Einstein, Veblen, Weyl, Alexander, and Morse. He remained at the Institute for the rest of his life, dying on 8 February 1957 in Washington, D.C., of cancer.

Von Neumann's contributions to mathematics, physics, and computer science are so extensive that any attempt at summarising them is bound to fail. In this paper, we shall emphasise his work on operator theory and on the mathematical foundations of quantum mechanics. Here we give a brief list of his other achievements: the axiomatisation of set theory (1925); the minimax theorem of game theory (1928); the theory of rings of operators, jointly with Murray (1936–1943); the ergodic theorem (1932); his work on the Manhattan Project; the design of the EDVAC and the von Neumann architecture for stored-program computers (1945); and the theory of self-replicating automata.

3.3 Von Neumann's Hilbert Space and the Spectral Theorem

Von Neumann's mathematical engagement with quantum mechanics began with the 1927 paper *Mathematische Begründung der Quantenmechanik* (*Mathematical Foundations of Quantum Mechanics*), *Göttinger Nachrichten*, pp. 1–57, written shortly before his arrival in Berlin. Two further fundamental papers followed in 1927: *Wahrscheinlichkeitstheoretischer Aufbau der Quantenmechanik* (*Probabilistic Construction of Quantum Mechanics*) and *Thermodynamik quantenmechanischer Gesamtheiten* (*Thermodynamics of Quantum-Mechanical Systems*). The 1929 paper *Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren* (*General Eigenvalue Theory of Hermitian Functional Operators*), *Mathematische Annalen* 102, pp. 49–131, contained the first rigorous proof of the spectral theorem for unbounded self-adjoint operators on Hilbert space.

Then, in a series of related papers from 1929–1932, von Neumann completed the abstract theory. The probably most important achievement of this period was the 1932 monograph *Mathematische Grundlagen der Quantenmechanik* (*Mathematical Foundations of Quantum Mechanics*), published by Springer in Berlin. The book summarised and unified von Neumann's earlier work on quantum mechanics. It established the modern formalism: **states are unit vectors, or more generally, density matrices** in a separable Hilbert space; **observables are self-adjoint operators**; and **time evolution is governed by the Schrödinger equation** $i\hbar \partial_t \psi = H\psi$.

In this paper we shall focus on the **spectral theorem for unbounded self-adjoint operators**. To state it, we must first define the relevant objects.

Definition 3.1. A **linear operator** A on a Hilbert space H is a pair consisting of a (linear) subspace $\text{dom}(A) \subset H$, called the **domain** of A , and a linear map $A : \text{dom}(A) \rightarrow H$.

The **adjoint** A^* is defined on the set of all $y \in H$ for which the linear functional $x \mapsto \langle Ax, y \rangle$, defined on $\text{dom}(A)$, is continuous; for such y , A^*y is the unique vector with $\langle Ax, y \rangle = \langle x, A^*y \rangle$ for all $x \in \text{dom}(A)$.

The operator A is called **symmetric** if $\text{dom}(A) \subset \text{dom}(A^*)$ and $A^*x = Ax$ on $\text{dom}(A)$, and **self-adjoint** if $\text{dom}(A) = \text{dom}(A^*)$ and $A^* = A$.

The distinction between symmetric and self-adjoint operators, which is invisible in the bounded case, is actually fundamental in the unbounded case.

Theorem 3.2 (Spectral theorem for unbounded self-adjoint operators). *Let A be a self-adjoint operator on a separable Hilbert space H . Then there exists a unique projection-valued*

measure E on the Borel sets of $\mathbb{R}$ (the so-called spectral measure of A) such that

$$A = \int_{\mathbb{R}} \lambda dE(\lambda),$$

in the sense that for every $x \in \text{dom}(A)$,

$$\langle Ax, x \rangle = \int_{\mathbb{R}} \lambda d\langle E(\lambda)x, x \rangle.$$

The support of E is the spectrum $\sigma(A) \subset \mathbb{R}$.

The proof, in the form von Neumann gave it, proceeds via the **Cayley transform**: the operator $U = (A - i)(A + i)^{-1}$ is unitary, and the spectral theorem for unitary operators can be reduced to the spectral theorem for self-adjoint bounded operators by **writing the unitary as $\exp(iB)$ for a bounded self-adjoint B** .

The core of this idea is the construction, for any bounded self-adjoint operator B , of the corresponding spectral family $\{E_\lambda\}_{\lambda \in \mathbb{R}}$, an increasing right-continuous family of orthogonal projections with limits 0 and I . The construction can be carried out either by approximating B by polynomials and using continuity, or, as in Stone's parallel approach, by a more direct argument based on the theory of resolutions of the identity.

The unbounded spectral theorem has great importance in quantum mechanics. For instance, **position and momentum operators** on $L^2(\mathbb{R})$, defined by $Q\psi(x) = x\psi(x)$ and $P\psi(x) = -i\hbar\psi'(x)$, are unbounded; their spectra are all of $\mathbb{R}$; and they have no eigenvectors in L^2 . Thus, the basic structure of quantum mechanics could be made entirely rigorous.

3.4 Marshall Stone and the One-Parameter Unitary Groups

Working independently of von Neumann but also in close communication, the American mathematician **Marshall Harvey Stone** obtained essentially the same results from a slightly different perspective.

3.4.1 Life of Marshall Stone

Stone was born on 8 April 1903 in New York City. His father Harlan Fiske Stone later became Chief Justice of the United States from 1941 to 1946.

Marshall was expected to follow his father into the law, but instead he became enchanted by mathematics during his Harvard undergraduate years. He took his A.B. in 1922 and his Ph.D. in 1926, both at Harvard, with a thesis directed by George David

Birkhoff on textit Ordinary linear homogeneous differential equations of order n and the related expansion problems.

After holding positions at Columbia, Harvard, and Yale, Stone settled at Harvard as full professor in 1937. In 1946 he accepted the chair of the Department of Mathematics at the University of Chicago, where he carried out a celebrated reform of the department, which lead to the so-called *Stone Age* of Chicago mathematics. He hired André Weil, S. S. Chern, Saunders Mac Lane, A. A. Albert, and many other excellent mathematicians.

Stone died on 9 January 1989 in Madras, India, where he had retired.

3.4.2 Contributions

Stone's contributions to mathematics include the **Stone–Weierstrass theorem** in approximation theory, **the Stone representation theorem** for Boolean algebras giving rise to Stone duality and an entire branch of category theory, and the **Stone–Čech compactification**, in addition to his work in spectral theory.

In this paper we state two results related to our topic, in this and the next subsection.

The first is what is now called **Stone's theorem on one-parameter unitary groups**, established in his paper *On one-parameter unitary groups in Hilbert space*, *Annals of Mathematics* 33 (1932), pp. 643–648.

Theorem 3.3 (Stone, 1932). *Let $\{U_t\}_{t \in \mathbb{R}}$ be a strongly continuous one-parameter family of unitary operators on a Hilbert space H , satisfying $U_{s+t} = U_s U_t$ for all $s, t \in \mathbb{R}$. Then there exists a unique self-adjoint operator A on H such that $U_t = \exp(itA)$ for all $t \in \mathbb{R}$. Conversely, every self-adjoint operator A determines, via this formula, a strongly continuous one-parameter unitary group.*

This theorem shows, in completely abstract operator-theoretic language, the structure of time evolution in quantum mechanics: a system whose state evolves according to a strongly continuous unitary symmetry of time is governed by a unique self-adjoint operator, the **generator** of the dynamics.

In the quantum-mechanical case, the generator is the Hamiltonian up to a factor of $-1/\hbar$.

The proof of the theorem makes essential use of the spectral theorem: given the unitary group U_t , one constructs the generator A by differentiating U_t at $t = 0$ on a dense subspace, and one verifies that the resulting operator is self-adjoint. Conversely, given the self-adjoint generator A , one defines $U_t = \exp(itA)$ by functional calculus.

The same year, 1932, Stone published his book-length monograph *Linear Transformations in Hilbert Space and Their Applications to Analysis*, AMS Colloquium Publications

volume 15, 622 pages. The book was a comprehensive summary of the theory of self-adjoint operators on Hilbert space, including the spectral theorem in its full generality (both bounded and unbounded cases), the perturbation theory of self-adjoint operators, and applications to ordinary differential equations and the theory of Sturm–Liouville problem. The book is of great significance even nowadays.

3.5 The Stone–von Neumann Uniqueness Theorem

A second fundamental result that emerged from the von Neumann–Stone collaboration is what is now called the **Stone–von Neumann uniqueness theorem**. This addresses the question of how Heisenberg’s matrix mechanics and Schrödinger’s wave mechanics can be the same theory: it provides the rigorous statement that, up to unitary equivalence, the canonical commutation relation $[Q, P] = i\hbar$ admits essentially only one irreducible representation by self-adjoint operators on a separable Hilbert space.

Theorem 3.4 (Stone–von Neumann). *Let $\{U_t\}_{t \in \mathbb{R}}$ and $\{V_s\}_{s \in \mathbb{R}}$ be strongly continuous one-parameter unitary groups on a separable Hilbert space H satisfying the Weyl relations*

$$U_t V_s = e^{ist} V_s U_t.$$

If the representation is irreducible, then it is unitarily equivalent to the standard Schrödinger representation on $L^2(\mathbb{R})$, with $U_t = e^{itQ}$ and $V_s = e^{isP}$.

The theorem is fundamental because it shows that, **in the framework of separable Hilbert space, Heisenberg’s matrix mechanics and Schrödinger’s wave mechanics are not merely physically equivalent, as Schrödinger had shown in 1926, but also mathematically equivalent in the strongest possible sense**. They are realizations of the same abstract algebraic structure, distinguished only by the choice of representation.

In 1930, Stone announced the result in his paper *Linear transformations in Hilbert space III: Operational methods and group theory*, in the *Proceedings of the National Academy of Sciences*, and in 1931 von Neumann gave a complete proof in his paper *Die Eindeutigkeit der Schrödingerschen Operatoren* (*Mathematische Annalen* 104, pp. 570–578).

The Stone–von Neumann theorem fails for systems with infinitely many degrees of freedom, where inequivalent representations of the canonical commutation relations exist. This is why quantum field theory and the theory of phase transitions are complicated. The classification of such inequivalent representations is one of the central problems in

mathematical physics, and it ties spectral theory directly to the theory of operator algebras and to non-commutative geometry.

3.6 The Hille–Yosida Theorem and Semigroups

Stone’s theorem characterises the unbounded self-adjoint operators that arise as generators of one-parameter unitary groups. The corresponding generalization to non-unitary semigroups is the **Hille–Yosida theorem**, established independently by **Einar Hille** and **Kosaku Yosida** in the late 1940s.

First, we give some definitions.

A **strongly continuous semigroup** of bounded operators on a Banach space X is a family $\{T_t\}_{t \geq 0}$ such that $T_0 = I$, $T_{s+t} = T_s T_t$, and $\lim_{t \downarrow 0} T_t x = x$ for all $x \in X$.

The **infinitesimal generator** is the operator $A = \lim_{t \downarrow 0} (T_t - I)/t$, defined on the dense subspace where the limit exists.

The Hille–Yosida theorem characterises precisely those operators A that arise in this way: **A must be densely defined and closed, and its resolvent $(A - \lambda I)^{-1}$ must satisfy specific norm bounds for all sufficiently large λ .** The theorem extends the spectral analysis of self-adjoint operators to a much wider class of evolution equations, including parabolic problems on Banach spaces and the Markov processes of probability theory.

Now we give the result explicitly.

Theorem 3.5 (Hille–Yosida). *1. Let $\{T_t\}_{t \geq 0}$ be a strongly continuous semigroup of bounded operators on a Banach space X . There exist constants M, ω such that $\|T_t\| \leq M e^{\omega t}$ for all $t \geq 0$.*

2. Conversely, a closed densely-defined operator A on X generates such a semigroup if and only if every $\lambda > \omega$ lies in the resolvent set of A , and the resolvent satisfies $\|(\lambda - A)^{-n}\| \leq M(\lambda - \omega)^{-n}$ for all $n \in \mathbb{N}$.

The theory of operator semigroups, after its formulation by Hille and Yosida, was developed extensively in the 1950s and 1960s by Phillips, Lumer, Trotter, Kato, and others. It has become the standard abstract framework for **time-dependent partial differential equations**: linear evolution equations $u'(t) = Au(t)$ with appropriate growth conditions on A . They are studied via the semigroup e^{tA} of operators that they generate, the regularity of solutions is analysed in terms of fractional powers of $-A$, and questions of stability or asymptotic behaviour are reformulated as spectral questions about A .

This result is also a key extension in spectral theory. In many physical and mathematical applications one is interested in one-parameter semigroups of bounded operators that are not necessarily unitary: for example, the heat semigroup $e^{t\Delta}$ on $L^2(\mathbb{R}^n)$, which propagates initial data forwards in time.

In the last section, we shall return to this theorem briefly to talk about its use in PDE.

4 More Modern Perspectives: Functional Calculus and Operator Algebras

With the basic results given by spectral theorems for certain types of operators, and the work of many to gradually extend the range of operators whose spectra can be analyzed explicitly, mathematicians were finally able to make use of spectral theory in more advanced ways. This is what we will talk about in this section.

4.1 The Continuous Functional Calculus

A direct consequence of the spectral theorem for self-adjoint operators is the existence of a powerful machinery for constructing functions of operators, called the **functional calculus**.

In its simplest form, for a self-adjoint operator A on a Hilbert space H with spectral measure E , and for a bounded Borel measurable function $f : \mathbb{R} \rightarrow \mathbb{C}$, one defines

$$f(A) = \int_{\mathbb{R}} f(\lambda) dE(\lambda).$$

This integral can be made precise as a strong limit of sums of the form $\sum_k f(\lambda_k) (E(\lambda_{k+1}) - E(\lambda_k))$ over partitions of the spectrum of A . The resulting map $f \mapsto f(A)$, from bounded Borel functions to bounded operators on H , is a $*$ -homomorphism of $*$ -algebras with the natural properties one would expect.

For example, taking $f(\lambda) = e^{it\lambda}$ gives $f(A) = e^{itA}$, recovering the unitary group of Stone's theorem; taking $f(\lambda) = \mathbf{1}_{(-\infty, c]}(\lambda)$ gives the spectral projection $E((-\infty, c])$. For a Schrödinger operator $H = -\Delta + V$ with appropriate potential, the functional calculus produces the **heat semigroup** e^{-tH} , the **wave propagator** $\cos(t\sqrt{H})$, the **resolvent** $(H - z)^{-1}$, and the so-called **Strichartz operators** e^{itH} . All these objects play essential roles in modern partial differential equations and mathematical physics.

A crucial subtlety appears for unbounded A and unbounded f : the operator $f(A)$ is then itself unbounded, with a domain that depends on f . The price one pays for this flexibility is that one must carefully track domains, and various manipulations valid in the bounded case require justification in the unbounded case.

The functional calculus has played a particularly important role in the modern study of Schrödinger operators. Most of the deep results of mathematical scattering theory—the existence and completeness of wave operators, the absolute continuity of the spectrum on a given interval, the asymptotic behaviour of e^{itH} as $t \rightarrow \infty$ —are statements about

specific elements of the functional calculus of H , proved by harmonic-analytic methods.

4.2 The Holomorphic and Riesz–Dunford Functional Calculus

For non-self-adjoint operators, the spectral theorem in its full strength is not available, but a more limited functional calculus can still be defined for holomorphic functions. Extension in definition is still on progress, using complex analysis.

The **Riesz–Dunford holomorphic functional calculus** associates, to any closed operator A on a Banach space X and any function f holomorphic on a neighbourhood of the spectrum of A , a bounded operator $f(A)$ defined by the contour integral

$$f(A) = \frac{1}{2\pi i} \oint_{\Gamma} f(z)(zI - A)^{-1} dz,$$

where Γ is a closed contour enclosing $\sigma(A)$ in the domain of holomorphy of f .

This calculus reduces to ordinary substitution when f is a polynomial, gives the operator exponential when $f(z) = e^{tz}$, and extends to a homomorphism of the algebra of germs of holomorphic functions on $\sigma(A)$ into the algebra of bounded operators on X . It was developed in its definitive form by Frédéric Riesz and Nelson Dunford in the 1930s and 1940s.

Comparison: Holomorphic functional calculus VS Borel calculus

When A is self-adjoint, the holomorphic functional calculus and the Borel functional calculus agree on functions that are holomorphic in a neighbourhood of $\sigma(A)$, but the Borel calculus is much richer: it makes sense even for measurable functions like indicator functions of Borel sets, which are needed for the construction of spectral projections. For general operators on a Banach space, however, the holomorphic calculus may be the most one can hope for.

A modern refinement

A. McIntosh, A. Pryde, and others gave the following refinement of the **H^∞ -functional calculus** for sectorial operators on Banach spaces, which extends the holomorphic calculus to bounded holomorphic functions on a sector.

4.3 Schatten Classes and the Theory of Operator Ideals

Between the bounded and the compact operators on a Hilbert space H lies a family of intermediate ideals, the so-called **Schatten–von Neumann classes** $S_p(H)$ for $1 \leq p < \infty$.

In particular, for $p = 2$, $S_2(H)$ is the class of **Hilbert–Schmidt operators**, those compact operators T for which the singular values $(s_n(T))$ are square-summable; For

$p = 1$, $S_1(H)$ is the class of **trace-class operators**, on which the trace functional $\text{Tr } T = \sum_n \langle T e_n, e_n \rangle$ is well-defined and independent of the choice of orthonormal basis.

The space $S_p(H)$ equipped with the norm $\|T\|_p = (\sum_n s_n(T)^p)^{1/p}$ is a two-sided ideal in $B(H)$ and a Banach space in its own right. Moreover, S_2 is a Hilbert space, with inner product $\langle S, T \rangle_{S_2} = \text{Tr}(T^* S)$.

These spaces play a crucial role in many parts of analysis.

The Hilbert–Schmidt operators on $L^2(\mathbb{R}^n)$ are precisely the integral operators with square-integrable kernels: $T_K f(x) = \int K(x, y) f(y) dy$ is Hilbert–Schmidt if and only if $K \in L^2(\mathbb{R}^n \times \mathbb{R}^n)$, and $\|T_K\|_{S_2} = \|K\|_{L^2}$.

As another example, trace-class operators play the role of “finite-volume” operators in non-commutative integration theory: the trace, originally defined by R. Schatten and J. von Neumann in their 1946 paper *The cross-space of linear transformations*, generalises the integral, and the duality $S_1^* = B(H)$ parallels the Riesz representation theorem for measures.

The theory of these classes was extensively developed in subsequent decades by Schatten, Calkin, Gohberg, Kreĭn, and many others, and it provides the foundation for the construction of regularised determinants, zeta functions of operators, and related objects of mathematical physics.

A particularly elegant application is the so-called **Lidskii trace theorem** (V. B. Lidskiĭ, 1959), which asserts that for any trace-class operator T on a Hilbert space, the operator-theoretic trace coincides with the spectral trace:

$$\text{Tr } T = \sum_n \lambda_n(T),$$

where the sum is over the (in general complex) eigenvalues of T , counted with algebraic multiplicities. This is a generalization of finite-dimensional linear algebra to the infinite-dimensional setting.

4.4 Murray–von Neumann Algebras: Rings of Operators

Beginning in 1936, von Neumann and Francis Joseph Murray studied what they called **rings of operators**, or **von Neumann algebras**.

The motivation came from quantum mechanics, where the natural object is not really a single self-adjoint operator but rather an entire algebra of observables. In from quantum statistical mechanics, where the system has infinitely many degrees of freedom, this is also true.

The series of papers *On rings of operators*, by Murray and von Neumann, appeared in *Annals of Mathematics* between 1936 and 1943, with a fourth paper by von Neumann alone.

4.4.1 The von Neumann algebra

Now, we shall go through the results.

A **von Neumann algebra** is a unital $*$ -subalgebra M of the algebra $B(H)$ of bounded operators on a Hilbert space H , closed in the weak operator topology.

Equivalently, by von Neumann’s celebrated double commutant theorem, M coincides with its **bicommutant**: $M = M''$, where $M' = \{T \in B(H) : TS = ST \text{ for all } S \in M\}$ is the **commutant**.

Von Neumann algebras admit a profound classification theory based on the structure of their projections.

A **factor** is a von Neumann algebra whose centre consists only of multiples of the identity. Murray and von Neumann **classified factors by the dimension or trace defined on projections, into types I, II, and III, with subtypes I_n , I_∞ , II_1 , II_∞ , and III.**

Roughly speaking, type I factors are isomorphic to $B(K)$ for some Hilbert space K , and they correspond to “**classical**” **quantum-mechanical settings**; type II factors arose in connection with infinite tensor products and **quantum statistical mechanics**, and admit a continuous but not integer-valued dimension function; and type III factors were initially mysterious but turned out to play a central role in **quantum field theory** and in the work of Connes on hyperfinite factors.

4.4.2 More extensions

The theory of von Neumann algebras was extended in subsequent decades by **Sakai, Tomita, Takesaki, Connes**, and others.

One of the decisive breakthroughs were the **Tomita–Takesaki theory** of modular automorphisms in 1970, which **provides every von Neumann algebra with a canonical one-parameter group of automorphisms**,

Another significant finding was **Connes’s classification** of injective factors of type III, which won him the Fields Medal in 1982. Connes showed that, **up to isomorphism, there is a unique injective factor of each type II_1 , II_∞ , III_λ for $0 \leq \lambda < 1$.** The factor of type III_1 had to wait for a 1987 result of **Haagerup**.

These results revealed the rigid structure of operator algebras in an algebraic and structural language.

4.4.3 Details and proof

The Tomita–Takesaki construction

Given a von Neumann algebra $M \subset B(H)$ and a cyclic and separating vector $\Omega \in H$ (cyclic means $M\Omega$ is dense in H ; separating means $a\Omega = 0$ implies $a = 0$ for $a \in M$), one defines an antilinear operator S on $M\Omega$ by $S(a\Omega) = a^*\Omega$. The closure $\bar{S}$ admits a polar decomposition $\bar{S} = J\Delta^{1/2}$, where J is an antiunitary involution and Δ is a positive (in general unbounded) self-adjoint operator. The remarkable result of M. Tomita, refined and made widely accessible by M. Takesaki in his 1970 lecture notes *Tomita's theory of modular Hilbert algebras and its applications*, is that the modular automorphism group $\sigma_t(a) = \Delta^{it}a\Delta^{-it}$ maps M to itself, and $JMJ = M'$. Every von Neumann algebra (with cyclic and separating vector) thus carries a canonical one-parameter group of automorphisms, intrinsic to the algebra and not depending on a choice of representation. This was a genuinely revolutionary insight: it showed that non-commutative measure theory has a built-in dynamics that has no counterpart in the classical commutative theory. Connes's subsequent realisation that the orbit of the modular flow under inner automorphisms is an invariant of the algebra (the so-called *flow of weights*) led directly to his classification of type III factors via the parameter $\lambda \in [0, 1]$, where $\lambda = 1$ corresponds to the most homogeneous case.

Conne's classification

Connes's 1973 paper *Une classification des facteurs de type III*, in *Annales scientifiques de l'École Normale Supérieure*, established the classification of type III factors into the subtypes III_λ for $\lambda \in [0, 1]$, based on Connes's spectral invariant $S(M)$, defined as the intersection of the spectra of all modular operators Δ_ϕ as ϕ ranges over the faithful normal states on M . For an injective (or hyperfinite, the two notions agree by Connes's celebrated 1976 theorem) factor, this single invariant determines the algebra up to isomorphism. The entire classification can be summarised: there is a unique hyperfinite factor of each of the types I_n ($n \in \{1, 2, \dots\} \cup \{\infty\}$), II_1 , II_∞ , III_λ for $0 \leq \lambda < 1$, and III_1 (the last by Haagerup). This rigidity result has no parallel anywhere else in mathematics: it asserts that, in the world of injective von Neumann factors, the type completely determines the algebra.

4.5 C*-Algebras and the Gelfand–Naimark Theorem

A parallel development, also rooted in the 1940s, is the theory of **C*-algebras**. A C*-algebra is a Banach *-algebra A over the complex numbers in which the norm satisfies the C*-identity

$$\|a^*a\| = \|a\|^2 \quad \text{for all } a \in A.$$

The principal commutative and non-commutative cases are the **algebra $C_0(X)$ of continuous complex-valued functions vanishing at infinity** on a locally compact Hausdorff space X and the algebra $B(H)$ of **bounded operators on a Hilbert space H** (the principal non-commutative case), or any norm-closed *-subalgebra thereof.

4.5.1 The 1943 paper

In 1943, **Israel Gelfand** and Mark Naimark stated in their paper *On the embedding of normed rings into the ring of operators in Hilbert space*, *Matematicheskii Sbornik* 12, pp. 197–213, is that these are essentially the only examples.

Theorem 4.1 (Commutative Gelfand–Naimark). *Let A be a commutative unital C*-algebra. Then there is a compact Hausdorff space X (the Gelfand spectrum of A) and an isometric *-isomorphism $A \cong C(X)$.*

The idea is simple. X is constructed as the set of non-zero algebra homomorphisms $A \rightarrow \mathbb{C}$, equipped with the weak-* topology. The Gelfand transform sends $a \in A$ to the function $\hat{a} : X \rightarrow \mathbb{C}$ defined by $\hat{a}(\chi) = \chi(a)$. The C*-identity is precisely what is needed to ensure that the Gelfand transform is an isometry.

Theorem 4.2 (Non-commutative Gelfand–Naimark). *Every C*-algebra A admits a faithful isometric *-representation as a norm-closed *-subalgebra of $B(H)$ for some Hilbert space H .*

The proof of the non-commutative theorem is via the **GNS construction (Gelfand–Naimark–Segal)**, introduced in its modern form by Segal in 1947. Given any state $\phi : A \rightarrow \mathbb{C}$ a positive linear functional with $\phi(1) = 1$, one constructs an inner product on A by $\langle a, b \rangle_\phi = \phi(b^*a)$, and passes to the quotient by the null vectors. The resulting Hilbert space carries a natural representation of A . Taking the direct sum over a sufficiently rich family of states yields a faithful representation.

The Gelfand–Naimark theorem establishes a categorical **duality** between commutative C*-algebras and locally compact Hausdorff spaces: **any continuous map $f : X \rightarrow Y$ of compact Hausdorff spaces gives rise to a *-homomorphism $f^* : C(Y) \rightarrow C(X)$**

by $f^*(g) = g \circ f$; conversely, any $*$ -homomorphism between commutative unital C^* -algebras arises in this way.

Compact Hausdorff spaces and continuous maps are thus equivalent, as a category, to commutative unital C^* -algebras and $*$ -homomorphisms (with arrows reversed).

4.5.2 Noncommutative algebra and geometry

The duality mentioned above is the conceptual seed of **non-commutative geometry**. If commutative C^* -algebras are the same as topological spaces, then non-commutative C^* -algebras are, by analogy, “non-commutative topological spaces”.

The programme of Alain Connes, beginning with his 1985 monograph and continuing through his 1994 book *Noncommutative Geometry*, has been to make this analogy precise. He tried to develop machinery of differential and Riemannian geometry: K-theory, cyclic cohomology, spectral triples, the local index formula, for non-commutative C^* -algebras.

Spectral theory plays a central role here: a **spectral triple** (A, H, D) consists of a C^* -algebra A acting on a Hilbert space H and a self-adjoint Dirac-type operator D such that $[D, a]$ is bounded for a in a dense subalgebra. **The spectral data of D , i.e. the spectral zeta function $\zeta_D(s) = \text{Tr}(|D|^{-s})$, encode geometric information about the underlying non-commutative manifold.**

4.5.3 Relation in between: K-theory

The key technical tool that links spectral theory to the topological invariants of non-commutative spaces is **K-theory**.

For a C^* -algebra A , one defines abelian groups $K_0(A)$ and $K_1(A)$ that generalise the topological K-theory of compact Hausdorff spaces. The Bott periodicity theorem of topological K-theory remains valid: $K_n(A) \cong K_{n+2}(A)$.

Concrete classes in $K_0(A)$ are represented by projections in matrix algebras over A , modulo a stable equivalence relation; classes in $K_1(A)$ are represented by unitaries in matrix algebras over A , again modulo equivalence.

The fundamental **index map** associates to a Fredholm operator T , i.e. an operator with finite-dimensional kernel and cokernel, the integer $\text{index}(T) = \dim \ker T - \dim \text{coker } T$.

More generally, the **Atiyah–Singer index theorem** in 1963, reformulated in K-theoretic terms, gives a topological formula for the index of an elliptic differential operator on a compact manifold.

4.6 Banach Algebras and Gelfand's Spectral Theory

A precursor to the C^* -algebra theory was Gelfand's general theory of **commutative Banach algebras**, developed in his 1939–1941 papers.

Gelfand showed that every commutative unital Banach algebra A has a non-empty maximal ideal space $\Delta(A)$ = the set of non-zero algebra homomorphisms to $\mathbb{C}$, which is a compact Hausdorff space. He also showed that any such A had a **Gelfand transform** $A \rightarrow C(\Delta(A))$. In particular, when A is the algebra $L^1(G)$ for a locally compact abelian group G , the Gelfand spectrum is the Pontryagin dual $\hat{G}$, and the Gelfand transform is the classical Fourier transform.

This finding unified harmonic analysis to a certain extent. The **Wiener tauberian theorem**, for example, becomes the statement that an ideal $I \subset L^1(\mathbb{R})$ is dense if and only if the Fourier transforms of its elements have no common zero on $\hat{\mathbb{R}} = \mathbb{R}$ —which is, in turn, a consequence of the Stone–Weierstrass theorem applied to $C_0(\hat{\mathbb{R}})$.

Israel Moiseevich Gelfand, born in 1913, was a Soviet and later American mathematician, whose work spanned representation theory, integral geometry, generalised functions, hypergeometric functions, and biology.

Born in Okny to a family of modest means, he never completed secondary school and arrived in Moscow at the age of sixteen working as a porter at the Lenin Library. He attended Moscow State University as an external student, was admitted to graduate school under Kolmogorov in 1932, and defended his doctoral thesis on Banach algebras in 1940.

Gelfand's Moscow seminar, which ran weekly from 1943 until his emigration in 1989, was for nearly half a century one of the great institutions of mathematics, attended by generations of Soviet mathematicians and producing dozens of distinguished students, including Naimark, Kirillov, Berezin, Graev, and many others.

Gelfand emigrated to the United States in 1989, joining Rutgers University, where he remained until his death in 2009 at the age of 96.

Mark Aronovich Naimark, born in 1909, was Gelfand's principal collaborator on the C^* -algebra theorem. He had a similarly Soviet-mathematical career.

Born in Odessa, Naimark studied at Odessa State University and then at the Steklov Institute in Moscow. His doctoral thesis in 1941 was on the representation theory of locally compact groups, and his post-war work, much of it joint with Gelfand, established many of the foundational results of operator-algebra theory.

4.7 Toward Non-Commutative Geometry

Connes's programme of non-commutative geometry is in many ways the natural conclusion of the line of thought initiated by Sturm and Liouville two centuries earlier.

The original aim was to understand the spectrum of a differential operator on a manifold, Nowadays, the aim has turned to using that spectrum and other general spectral invariants of operators on Hilbert space to define the manifold itself. As is said in the slogan **Spectrum defines geometry**.

For instance in particle physics, Connes has shown that the Lagrangian of the standard model can be derived from a particular spectral triple involving a finite-dimensional non-commutative algebra encoding the internal symmetries of the standard model.

4.7.1 A concrete case

To give some sense of how this programme works in practice: a classical Riemannian manifold (M, g) gives rise to a spectral triple $(C^\infty(M), L^2(M, S), D)$, where $C^\infty(M)$ is the algebra of smooth functions on M , $L^2(M, S)$ is the Hilbert space of L^2 -spinors, and D is the Dirac operator.

On the other hand, from this triple, one can recover the metric g by **Connes's distance formula**

$$d(x, y) = \sup\{|f(x) - f(y)| : f \in C^\infty(M), \|[D, f]\| \leq 1\},$$

which expresses the metric as a supremum over Lipschitz functions, the constraint $\|[D, f]\| \leq 1$ being the **operator-theoretic version of the Lipschitz constraint** $\|\nabla f\|_\infty \leq 1$. One can also **recover the dimension from the asymptotic behaviour of the eigenvalues of D , the volume from the leading coefficient of the Weyl asymptotics, and the integrated scalar curvature from the second coefficient of the heat-kernel expansion**.

Connes' work showed that **all of this works in the non-commutative case**, where the algebra is no longer commutative and there is no underlying manifold; the spectral data of the operator D carries the geometric information directly.

4.7.2 Connes' work

In 1996, Connes and Chamseddine formulated the **spectral action principle**, and provided a guiding heuristic for constructing physical Lagrangians from spectral triples.

The action is taken to be $S = \text{Tr}(\chi(D/\Lambda))$, where χ is a smooth even cut-off function and Λ is an energy scale. Expanding asymptotically in $\Lambda \rightarrow \infty$ using heat-kernel methods yields a Lagrangian that, for an appropriate choice of spectral triple involving a finite-dimensional algebra describing the standard-model gauge group and matter content, reproduces the bosonic part of the standard-model Lagrangian. This is an undeniably elegant result.

5 Applications of Spectral Theory

In this section we consider some of the most important applications of spectral theory in modern mathematics and physics, or more specifically, its influence in **harmonic analysis, PDE solving, probability theory and stochastic processes, numerical analysis, quantum physics and differential geometry**.

5.1 Fourier Analysis and Harmonic Analysis

Recall that the **Fourier transform** $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ which is a unitary operator:

$$(\mathcal{F}\phi)(\xi) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \phi(x) e^{-ix \cdot \xi} dx,$$

From the spectral viewpoint, the Fourier transform is the diagonalization of the family of translation operators $T_a\phi(x) = \phi(x - a)$: the eigenfunctions of T_a in the sense of distribution are the exponentials $e^{ix \cdot \xi}$, and the Fourier transform is precisely the change of basis that puts every T_a into diagonal form. More abstractly, the Fourier transform is the Gelfand transform of the convolution algebra $L^1(\mathbb{R}^n)$ extended to L^2 by Plancherel's theorem.

Plancherel's theorem itself can be regarded as a spectral theorem for the differential operator $-i\nabla$ on $L^2(\mathbb{R}^n)$. The Fourier transform diagonalizes $-i\nabla$ into multiplication by ξ , and Plancherel's identity $\|\mathcal{F}\phi\|_{L^2} = \|\phi\|_{L^2}$ is the unitarity of this diagonalization.

The **Bochner theorem**, which characterizes the Fourier transforms of finite positive measures on $\mathbb{R}^n$ as the continuous positive-definite functions, is closely related. In modern language, it states that the Fourier transform sets up a bijection between positive measures on $\mathbb{R}^n$ and continuous positive-definite functions on $\mathbb{R}^n$, with the spectral measure of the convolution operator on $L^2(\mu)$ being precisely μ .

The **generalization of Fourier analysis to non-abelian groups** was carried out from the late 1940s onwards by Mackey, Mautner, and others, building on earlier work of Weyl, Peter, and von Neumann, by applying the abstract spectral theorem.

A central object of modern harmonic analysis is the **spectral multiplier theorem of Mihlin and Hörmander**, which gives sufficient conditions on a function m for the operator $m(\sqrt{-\Delta})$, defined via the Fourier transform, to be bounded on $L^p(\mathbb{R}^n)$. Such theorems are essential in the theory of singular integrals and pseudodifferential operators.

5.2 Ordinary and Partial Differential Equations

Spectral theory was originally developed to solve differential equations, and even after such a grand extension, it remains to be one of the most powerful general tools for solving equations.

We have already seen the application of Sturm–Liouville theory to second-order ODEs on a finite interval. The theory for higher-order ODEs or even PDEs on bounded or unbounded domains, is far more intricate but follows the same general pattern.

First, one identifies the differential operator as a self-adjoint operator on a Hilbert space with appropriate boundary conditions. Then one applies the spectral theorem to obtain a representation $A = \int \lambda dE(\lambda)$. Finally, one can reads off properties of solutions to the associated equations.

For elliptic operators on a bounded domain with reasonable boundary conditions, the resolvent is compact and the spectrum is discrete. One therefore obtains a complete orthonormal basis of eigenfunctions, and the eigenvalue problem becomes a generalised Fourier expansion. In 1911, Weyl proved in *Über die asymptotische Verteilung der Eigenwerte*) the celebrated asymptotic formula

$$N(\lambda) \sim \frac{\omega_n}{(2\pi)^n} \text{vol}(\Omega) \lambda^{n/2}$$

for the number of eigenvalues less than λ of the Laplacian on a domain $\Omega \subset \mathbb{R}^n$, where ω_n is the volume of the unit ball in $\mathbb{R}^n$.

5.2.1 Can one hear the shape of a drum?

Weyl’s law has been extended in many directions, leading to the famous **Can one hear the shape of a drum?** question of Mark Kac and the celebrated counterexamples of Gordon, Webb, and Wolpert.

The question raised by Kac in his 1966 paper *Can one hear the shape of a drum?* (*American Mathematical Monthly* 73, pp. 1–23) asks the following question: Given the spectrum of the Laplacian on a planar domain with Dirichlet boundary conditions, can one reconstruct the domain up to congruence?

Weyl’s law shows that the area can be heard. The **Kac–Pleijel formula** and the **McKean–Singer expansion** showed that the perimeter and even the topology can be heard.

For a long time the question of whether the full shape was determined remained open. In 1992, **Carolyn Gordon, David Webb, and Scott Wolpert** stated in their

paper *Isospectral plane domains and surfaces via Riemannian orbifolds* that the answer is negative. In other words, there **exist non-isometric planar domains with identical Laplace spectra**. Their construction uses **Sunada's method** of constructing isospectral pairs from finite groups acting on Riemannian manifolds, applied to a specific pair of polygonal domains. The pair of **GWW domains** has become a celebrated illustration of how **spectral theory both reveals and conceals geometric information**.

5.2.2 Evolution equations and semigroups

For evolution equations, the abstract framework of operator semigroups is the appropriate setting. In other words, the **spectral data of Δ governs the qualitative and quantitative behaviour of the solutions**. We give a few examples here.

1. The heat equation $\partial_t u = \Delta u$ on a Riemannian manifold M corresponds to the heat semigroup $e^{t\Delta}$, which is a strongly continuous semigroup of self-adjoint operators on $L^2(M)$ generated by the (negative of the) Laplace–Beltrami operator.
2. The wave equation $\partial_t^2 u = \Delta u$ gives rise to the unitary group generated by the matrix operator $\begin{pmatrix} 0 & I \\ \Delta & 0 \end{pmatrix}$ acting on a suitable Sobolev space.
3. The Schrödinger equation $i\partial_t \psi = -\Delta \psi$ gives the unitary group $e^{-it\Delta}$, to which Stone's theorem applies directly.

Abstract semigroup theory is the foundation of many modern studies on these semigroups, and therefore is also the base of the theory in PDE solving.

A remarkable application is the so-called **Trotter product formula**. Under appropriate hypotheses, the semigroup generated by a sum $A + B$ of (unbounded) operators as a limit of products of the semigroups generated by A and B :

$$e^{t(A+B)} = \lim_{n \rightarrow \infty} (e^{tA/n} e^{tB/n})^n.$$

This formula is the rigorous basis of the **Feynman path integral** for quantum-mechanical evolution: by approximating the Hamiltonian $H = -\Delta + V$ by a sum of kinetic and potential parts and using the Trotter formula, one represents the unitary evolution e^{-itH} as a limit of finite-dimensional integrals, which can be heuristically interpreted as an integral over paths in configuration space.

A closely related result, valid for the heat semigroup, is the **Feynman–Kac formula**: for $H = -\frac{1}{2}\Delta + V$ on $\mathbb{R}^n$ with a sufficiently regular potential V , the kernel of the heat

semigroup e^{-tH} admits the probabilistic representation

$$(e^{-tH}f)(x) = \mathbb{E}_x \left[\exp \left(- \int_0^t V(B_s) ds \right) f(B_t) \right],$$

where B_s is Brownian motion starting at x .

This formula, established by Mark Kac in 1949, is the deepest single bridge between spectral theory and probability theory: the spectral analysis of Schrödinger operators can be reduced to the analysis of certain Wiener path integrals, and conversely many subtle questions about Brownian functionals can be answered by spectral-theoretic computations.

The asymptotic behaviour of the **heat kernel** $K_t(x, y) = \langle x | e^{-tH} | y \rangle$ as $t \downarrow 0$ is a topic of independent interest. For example, for the **Laplace–Beltrami operator** on a compact Riemannian manifold M of dimension n , **Minakshisundaram and Pleijel** established the small-time expansion in 1949:

$$K_t(x, x) \sim \frac{1}{(4\pi t)^{n/2}} \sum_{k=0}^{\infty} a_k(x) t^k,$$

where the heat-kernel coefficients $a_k(x)$ are local invariants of the metric: $a_0 = 1$, $a_1 = R/6$ (where R is the scalar curvature), and so on. Integrating over M gives expansions of the heat trace $\text{Tr } e^{-tH}$ in terms of integrated curvature invariants, leading to a beautiful synthesis of spectral theory and Riemannian geometry..

5.3 Probability Theory and Stochastic Processes

We mention three of the most important links between spectral theory and probability theory.

The first is the spectral theory of **Markov processes**.

Given a Markov process $(X_t)_{t \geq 0}$ on a state space Ω , with transition kernel P_t , the operators $P_t f(x) = \mathbb{E}_x f(X_t)$ form a strongly continuous contraction semigroup on appropriate function spaces. The infinitesimal generator $\mathcal{L}$ of this semigroup encodes the dynamics of the process: for diffusions, $\mathcal{L}$ is a second-order differential operator; for jump processes, it is an integro-differential operator. **Spectral properties of $\mathcal{L}$ translate directly into probabilistic properties of the process.**

The second is the spectral theory of **random matrices**.

A random matrix is a matrix whose entries are random variables, thus one can study the joint distribution of its eigenvalues.

The modern subject was initiated by **Wigner’s 1955 papers** on the eigenvalue distribution of random Hermitian matrices, motivated by nuclear physics. Wigner showed that, for a wide class of random Hermitian matrices, the empirical spectral distribution converges, in the limit of large dimension, to the so-called semicircle law. The free probability theory of Voiculescu, beginning in the 1980s, provides a non-commutative probability framework in which the asymptotic spectral distribution of large random matrices arises naturally. There is also a connection to operator algebras.

The third example is the spectral theory of the **Schrödinger operator with random potential**, or the **Anderson model**.

Here one studies the operator $H_\omega = -\Delta + V_\omega$ on $\ell^2(\mathbb{Z}^d)$, where $V_\omega(x)$ is a random potential, typically i.i.d. random variables indexed by $x \in \mathbb{Z}^d$. The spectrum of H_ω is almost surely deterministic, but its detailed structure depends sensitively on the dimension d and the strength of the disorder.

This phenomenon of **Anderson localisation**, predicted by **P. W. Anderson** in 1958 and proved rigorously by **Fröhlich and Spencer** in 1983 and **Aizenman and Molchanov** in 1993, is one of the deepest results in the spectral theory of random operators: **In low dimensions, or for strong disorder in any dimension, the spectrum of H_ω is, almost surely, pure-point with exponentially decaying eigenfunctions.**

5.4 Numerical Analysis

The spectral theory of finite matrices, i.e. the theory of eigenvalues and eigenvectors of $n \times n$ matrices, is obviously a special case of the general operator-theoretic spectral theory, and it is the subject of an enormous literature in numerical linear algebra.

Indeed, many of the basic algorithms for computing eigenvalues—the QR algorithm, the Lanczos and Arnoldi iterations, the power method—all rely, at some level, on the spectral decomposition theorem, and their analysis uses classical results of **perturbation theory** for self-adjoint or normal operators.

Another is the use of spectral methods in numerical solution of partial differential equations, which we call **Spectral methods**.

This is introduced by **Steven Orszag** and others in the 1970s. They used the **eigenfunctions of an underlying differential operator**, such as Fourier or Chebyshev polynomials, **as a basis for the discretization**. In this way, for smooth problems they can achieve exponential convergence rates, which is far superior to the polynomial rates of finite-difference or finite-element methods.

5.5 Quantum Mechanics, Quantum Field Theory, and Beyond

We have already discussed in detail the role of spectral theory in the foundations of quantum mechanics, yet of course the subject has expanded in many directions afterwards. Here are some examples.

In **quantum field theory**, one works with operator-valued distributions on space-time, satisfying the Wightman axioms or the Haag–Kastler axioms. The local algebras of the Haag–Kastler framework are typically von Neumann algebras of type III, and their spectral theory connects to quantum statistical mechanics, the Tomita–Takesaki modular theory, and the theory of KMS states.

In **quantum information theory**, the spectral decomposition of density operators is fundamental. A state ρ on a finite-dimensional Hilbert space is a positive operator of trace one, and its **spectral decomposition** $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ gives the probabilities p_i of finding the system in various pure states $|\psi_i\rangle$. The **von Neumann entropy** $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$, which measures quantum uncertainty, is computed via the functional calculus.

5.6 Differential Geometry

In **semiclassical analysis** and **microlocal analysis**, the spectral theory of pseudodifferential and Fourier integral operators on manifolds plays a central role. The work of **Hörmander**, **Duistermaat**, **Sjöstrand**, **Helffer** and others has revealed deep connections between the asymptotic spectral properties of the Laplacian on a Riemannian manifold and the geometry of the underlying geodesic flow. The **Selberg trace formula**, the **Duistermaat–Guillemin theorem**, and the **Atiyah–Singer index theorem** are all spectral-theoretic in nature, and they form the bridge between spectral theory and global differential geometry.

Here, we shall give a brief introduction to two of these results.

The **Selberg trace formula** in 1956 originally formulated for hyperbolic surfaces $\Gamma \backslash \mathbb{H}$ where Γ is a Fuchsian group, asserts an identity between two summations: the sum over eigenvalues of the Laplacian on the surface, and the sum over conjugacy classes of Γ (geometrically, the sum over closed geodesics). It is a non-abelian analogue of the Poisson summation formula and the prototype of all subsequent trace formulas in number theory and representation theory.

For example, the **Duistermaat–Guillemin trace formula** in 1975 is a microlocal generalisation, valid for the Laplacian on any compact Riemannian manifold, that expresses the singularities of $\operatorname{Tr} \cos(t\sqrt{-\Delta})$ as a distribution on $\mathbb{R}$ in terms of the lengths of closed geodesics.

These formulas reveal, in concrete terms, the principle that **the spectrum of the Laplacian remembers the geometry of the underlying manifold**: the lengths of closed geodesics, the volumes of geodesically convex sets, the curvature integrals, and—in some cases—the manifold itself.

The **Atiyah–Singer index theorem** in 1963, proved by **Michael Atiyah** and **Isadore Singer**, gives a topological formula for the index of an elliptic differential operator P on a compact manifold M :

$$\text{index}(P) = \int_{T^*M} \text{ch}(\sigma(P)) \text{Td}(M),$$

where $\sigma(P)$ is the principal symbol of P , ch is the Chern character, and Td is the Todd class of the manifold.

Although this formula is about topology in the start, its proof and interpretation rely a lot on the spectral theory of pseudodifferential operators. The Atiyah–Singer theorem thus stands at the confluence of spectral theory, topology, and global analysis, and its many generalizations—to families of operators, to manifolds with boundary, to non-commutative spaces.

6 Conclusion

The introduction part of this paper has provided a quite comprehensive summary of its contents, so the conclusion part shall be about some naive thoughts of the author.

What's so surprising and interesting about spectral theory, is the fact that it can be extended in so many different ways, and can be continuously extended throughout such a wide span of time. Although extension is indeed one of the fundamental ideas of mathematical research, it is still uniquely remarkable when applied to this theory.

As a starter, I think physics, or reality, really made a huge difference. The in-born complexity of nature pushed scientists to decompose and classify the spectra of more irregular operators, which surprisingly influenced the subjects of algebra and geometry.

Secondly, I think, from a philosophical point of view, experts in history of science should focus on the love of men on patterns and simplifications. Spectral theory is a perfect combination of the two. Its elegance gave me a real start when I first met with Schmidt and Hilbert's spectral theorem. It is a real pleasure to find out how a wide category of objects can be so easily seen through. Sadly though men are creatures who fail to process multiple and high-dimensional experiments physically or mentally, the ability to do *decomposition* is an extremely precious gift for this species, gifted perhaps not so evenly among individuals.

Last but not least, the collaboration between mathematicians of different nationality and backgrounds, even during wartime, can be seen in the narrative and story-telling of this paper for multiple times, as is in most publications on history of maths. This is a touching feature of this subject, and of science, that has lasted till now. I believe all sincerely hope it will continue to show eternity.

Many thanks to fellow students and the respected lecturers who inspired me on writing this paper. History has had a great influence on the presence, and hence also the future. Hopefully, so does mathematics.

The underlying rules of logic is always the most fascinating among all findings of mankind.

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While writing this paper, the use of AI is mainly aimed to do brainstorm and to help the author gain a general picture of this topic before deeper studies. This was done while writing the outline, which was submitted a few months ago and the record of AI use back then cannot be found.

No AI was used to come up directly with literature in this paper, nor to make up references.

No AI was used to generate this paper based on works of others.